

CS 2601 Linear and Convex Optimization

Review

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Convex sets (1/2)

- definition

$$\mathbf{x}, \mathbf{y} \in C, \theta \in [0, 1] \implies \theta \mathbf{x} + \bar{\theta} \mathbf{y} \in C$$

- convex combination

$$\sum_{i=1}^k \theta_i \mathbf{x}_i, \quad \text{where } \theta_i \geq 0, \quad \sum_{i=1}^k \theta_i = 1$$

- convex hull of S , smallest convex set containing S , set of all convex combinations of points in S ,

$$\text{conv } S = \left\{ \sum_{i=1}^m \theta_i \mathbf{x}_i : m \in \mathbb{N}; \mathbf{x}_i \in S, \theta_i \geq 0, i = 1, \dots, m; \sum_{i=1}^m \theta_i = 1 \right\}$$

- examples: lines, rays, line segments, hyperplanes, half plane, affine space, polyhedron, norm ball, ellipsoid, simplex, positive semidefinite cone

Convex sets (2/2)

- convexity-preserving operations
 - ▶ intersection of convex sets
 - ▶ image/preimage of convex set under affine transformation
- projection onto closed convex set

$$\mathcal{P}_C(\mathbf{x}) = \operatorname{argmin}_{\mathbf{z} \in C} \|\mathbf{x} - \mathbf{z}\|_2 = \operatorname{argmin}_{\mathbf{z} \in C} \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|_2^2$$

- supporting hyperplane theorem
- separating hyperplane theorem

methods for proving convexity:

- definition
- convexity-preserving operations
- sublevel/superlevel set of convex/concave functions
- epigraph/hypograph of convex/concave functions

Convex functions (1/3)

- definition: f is convex if it has convex domain $\text{dom} f$, and

$$\mathbf{x}, \mathbf{y} \in \text{dom} f, \theta \in (0, 1) \implies f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$$

f is concave if $-f$ is convex.

- affine functions are the only functions that are both convex and concave.
- strict convexity

$$\mathbf{x} \neq \mathbf{y} \in \text{dom} f, \theta \in (0, 1) \implies f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) < \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$$

- strong convexity: f is m -strongly convex if $f(\mathbf{x}) - \frac{m}{2} \|\mathbf{x}\|_2^2$ is convex.
- examples: norm, negative entropy, log-sum-exp function, quadratic function with PSD quadratic term,...
- epigraph

$$\text{epi} f = \{(\mathbf{x}, y) : \mathbf{x} \in \text{dom} f, y \geq f(\mathbf{x})\}$$

f is a convex function iff $\text{epi} f$ is a convex set.

- sublevel sets of convex functions are convex

$$C_\alpha(f) = \{\mathbf{x} \in \text{dom} f : f(\mathbf{x}) \leq \alpha\}$$

Convex functions (2/3)

- zero-th order condition
 - ▶ restriction to any line is (strictly/strongly) convex
- first-order conditions
 - ▶ convexity

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}), \quad \forall \mathbf{x}, \mathbf{y} \in \text{dom} f$$

- ▶ strict convexity

$$f(\mathbf{y}) > f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}), \quad \forall \mathbf{x} \neq \mathbf{y} \in \text{dom} f$$

- ▶ m -strong convexity

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{m}{2}\|\mathbf{x} - \mathbf{y}\|_2^2, \quad \forall \mathbf{x}, \mathbf{y} \in \text{dom} f$$

- second-order conditions
 - ▶ convexity

$$\nabla^2 f(\mathbf{x}) \succeq \mathbf{O}, \quad \forall \mathbf{x} \in \text{dom} f$$

- ▶ strict convexity

$$\nabla^2 f(\mathbf{x}) \succ \mathbf{O}, \quad \forall \mathbf{x} \in \text{dom} f$$

- ▶ m -strong convexity

$$\nabla^2 f(\mathbf{x}) \succeq m\mathbf{I}, \quad \forall \mathbf{x} \in \text{dom} f$$

Convex functions (3/3)

- convexity preserving operations

- ▶ nonnegative combinations

$$f(\mathbf{x}) = \sum_{i=1}^m c_i f_i(\mathbf{x})$$

- ▶ composition with affine functions

$$f(\mathbf{x}) = g(\mathbf{A}\mathbf{x} + \mathbf{b})$$

- ▶ certain composition of monotonic convex/concave functions

$$f(\mathbf{x}) = h(g_1(\mathbf{x}), \dots, g_m(\mathbf{x}))$$

- ▶ pointwise maximum/supremum

$$f(\mathbf{x}) = \sup_{i \in I} f_i(\mathbf{x})$$

- ▶ partial minimization: for convex g and convex C ,

$$f(\mathbf{x}) = \inf_{\mathbf{y} \in C} g(\mathbf{x}, \mathbf{y})$$

Optimization problems

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{0}\end{array}$$

- domain $D = \text{dom} f \cap (\bigcap_i \text{dom } g_i) \cap (\bigcap_j \text{dom } h_j)$
- feasible set

$$X = \{\mathbf{x} \in D : \mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \mathbf{h}(\mathbf{x}) = \mathbf{0}\}$$

$\mathbf{x}$ is feasible if $\mathbf{x} \in X$

- $f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x})$ is the optimal value
- $\mathbf{x}^* \in X$ is a global minimum if $f^* = f(\mathbf{x}^*)$, or equivalently

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in X$$

- $\mathbf{x}^* \in X$ is a local minimum if for some $\delta > 0$,

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \quad \forall \mathbf{x} \in X \cap B(\mathbf{x}^*, \delta)$$

Convex optimization problems

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{0}\end{array}$$

- $f, \mathbf{g}$ are convex, $\mathbf{h} = \mathbf{Ax} - \mathbf{b}$ is affine.
- key property: local minima are global minima.
 - ▶ no assertion about existence; * some conditions for existence
 - ▶ no assertion about uniqueness; if f is strictly convex, solution is unique if exists.
- examples: LP, QP, QCQP, GP
- equivalent problems: informally, solution of one problem readily yields solution to the other
 - ▶ some simple transformation: changing variables, eliminating equality constraints, introducing slack variables, transforming objective/constraints,...
- be able to formulate simple convex optimization problems

Optimality conditions for smooth convex problems

- unconstrained problem

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

- constrained problem

$$\nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in X$$

- equality constrained problem: Lagrange condition

$$\begin{cases} \nabla f(\mathbf{x}^*) + \mathbf{A}^T \boldsymbol{\lambda}^* = \mathbf{0} \\ \mathbf{A} \mathbf{x}^* = \mathbf{b} \end{cases}$$

- inequality constrained problem: KKT conditions (necessary at regular point; sufficient)

- ▶ primal feasibility: $\mathbf{h}(\mathbf{x}^*) = \mathbf{0}, \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}$
- ▶ dual feasibility: $\boldsymbol{\mu}^* \geq \mathbf{0}$
- ▶ stationarity: $\nabla_x \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$
- ▶ complementary slackness: $\mu_j^* g_j(\mathbf{x}^*) = 0, \forall j$

Lagrange duality (1/2)

- general primal problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{0} \end{aligned}$$

- Lagrangian

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x})$$

- ▶ $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \leq f(\mathbf{x})$ for feasible $\mathbf{x}$ and $\boldsymbol{\mu} \geq \mathbf{0}$

- dual function

$$\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in D} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$$

- ▶ always concave
 - ▶ domain: $\{(\boldsymbol{\lambda}, \boldsymbol{\mu}) : \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty\}$
 - ▶ lower bound property $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) \leq \phi^* \leq f^* \leq f(\mathbf{x})$ for $\boldsymbol{\mu} \geq \mathbf{0}, \mathbf{x} \in X$

Lagrange duality (2/2)

- dual problem

$$\begin{aligned} \max_{\lambda, \mu} \quad & \phi(\lambda, \mu) \\ \text{s.t.} \quad & \mu \geq \mathbf{0} \end{aligned}$$

- ▶ always a convex optimization problem
 - ▶ dual LP that makes constraints explicit
- weak duality: $\phi^* \leq f^*$
 - ▶ optimal duality gap $f^* - \phi^*$
- strong duality: $\phi^* = f^*$
 - ▶ (refined) Slater's condition for convex problems
 - ▶ strong duality almost always holds for LP
- KKT conditions and strong duality for convex problems

KKT $\iff$ strong duality + primal optimality + dual optimality

Algorithms (1/2)

unconstrained problems

- smooth f
 - ▶ descent method: $\mathbf{x}_{k+1} = \mathbf{x}_k + t_k \mathbf{d}_k$
 - ▶ descent direction
 - ▶ negative gradient: $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$
 - ▶ Newton direction: $\mathbf{d}_k = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$
 - ▶ step size
 - ▶ constant
 - ▶ exact line search
 - ▶ backtracking line search (Armijo's rule)
 - ▶ * condition number
 - ▶ * convergence analysis
- * smooth f + nonsmooth h
 - ▶ proximal gradient descent

$$\mathbf{x}_{k+1} = \text{prox}_{t_k h}(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k))$$

$$\text{prox}_h(\mathbf{x}) = \underset{\mathbf{z}}{\operatorname{argmin}} \left\{ \frac{1}{2} \|\mathbf{z} - \mathbf{x}\|_2^2 + h(\mathbf{z}) \right\}$$

Algorithms (2/2)

constrained problems

- equality constraints
 - ▶ constraint elimination
 - ▶ Newton's method
 - ▶ KKT system for finding descent direction

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}_k) & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_k) \\ \mathbf{0} \end{bmatrix}$$

- * inequality constraints
 - ▶ projected gradient descent

$$\mathbf{x}_{k+1} = \mathcal{P}_X(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k))$$

- ▶ barrier method